

MI PANEL API

1. API Reference

1.1. Function Module API

Table 1-1

API name	Features
MI_PANEL_Init [#12-mi_panel_init]	Initialize PANEL
MI_PANEL_DeInit [#13-mi_panel_deinit]	to initialize PANEL
MI_PANEL_GetTotalNum [#14-mi_panel_gettotalnum]	Get the total number of screen parameters
MI_PANEL_SetPowerOn [#15-mi_panel_setpoweron]	Set screen power enable
MI_PANEL_GetPowerOn [#16-mi_panel_getpoweron]	Get screen power status
MI_PANEL_SetBacklight [#17-mi_panel_setbacklight]	Set Backlight Enable
MI_PANEL_GetBacklight [#18-mi_panel_getbacklight]	Get backlight enable status
MI_PANEL_SetBacklightLevel [#19-mi_panel_setbacklightlevel]	Set the backlight brightness value
MI_PANEL_GetBacklightLevel [#110-mi_panel_getbacklightlevel]	Get the backlight brightness value

MI_PANEL_SetSscConfig [#111-mi_panel_setsscconfig]	Set spread spectrum related information
MI_PANEL_SetMipiDsiConfig [#112-mi_panel_setmipidsiconfig]	Set MIPI related information
MI_PANEL_SetTimingConfig [#113-mi_panel_settimingconfig]	Set Timing related information
MI_PANEL_SetDrvCurrentConfig [#114-mi_panel_setdrvcurrentconfig]	Set drive capability
MI_PANEL_SetOutputPattern [#115-mi_panel_setoutputpattern]	Set PANEL output pattern
MI_PANEL_SetPanelParam [#116-mi_panel_setpanelparam]	Set up PANEL INFO
MI_PANEL_GetPanelParam [#117-mi_panel_getpanelparam]	Get current PANEL configuration information
MI_PANEL_GetTimingConfig [#118-mi_panel_gettimingconfig]	Get Timing related information
MI_PANEL_InitDev [#119-mi_panel_initdev]	Initialize the panel device
MI_PANEL_DeInitDev [#120-mi_panel_deinitdev]	Deinitialize the panel device

1.2. MI_PANEL_Init

- Features

Initialize the PANEL module.

- grammar

```
MI_S32 MI_PANEL_Init(MI_PANEL_IntfType_e eIntfType);
```

- Arguments

Table 1-2

parameter name	describe	input Output
eIntfType	PANEL interface type.	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_panel.h, mi_panel_datatype.h
 - Library files: libmi_panel.a / libmi_panel.so
- related topic
[MI_PANEL_DeInit](#) [#13-mi_panel_deinit]

1.3. MI_PANEL_DeInit

- Features

To initialize PANEL.

- grammar

```
MI_S32 MI_PANEL_DeInit(void);
```

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_panel.h, mi_panel_datatype.h
 - Library files: libmi_panel.a / libmi_panel.so

- Notice

Not currently supported.

1.4. MI_PANEL_GetTotalNum

- Features

Get the total number of screen parameters.

- grammar

```
MI_S32 MI_PANEL_GetTotalNum (MI_U32 * pu32TotalNum);
```

- Arguments

Table 1-5

parameter name	describe	input Output
pu32TotalNum	The pointer receives the number of screen parameters	output

- return value

- 0 success.
- If it is not 0, it fails. For details, refer to the [error code](#) [jump] .

- rely

- Header files: mi_panel.h, mi_panel_datatype.h
- Library files: libmi_panel.a / libmi_panel.so

- Notice

Not currently supported.

1.5. MI_PANEL_SetPowerOn

- Features

Set the screen power supply enable.

- grammar

```
MI_S32 MI_PANEL_SetPowerOn(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_PowerConfig_t *pstPowerCfg);
```

- Arguments

Table 1-6

parameter name	describe	input Output
eIntfType	Interface Type	enter
pstPowerCfg	Screen power enable status	enter

- return value

- 0 success.
- 非0 失败，[错误码](#) [#jump]。

- 依赖

- 头文件：mi_panel.h、mi_panel_datatype.h
- 库文件：libmi_panel.a / libmi_panel.so

- 注意

- 设置屏供电使能之前需要初始化PANEL。
- 目前不支持，建议app直接控制GPIO。

- 相关主题

[MI_PANEL_GetPowerOn](#) [#16-mi_panel_getpoweron]

1.6. MI_PANEL_GetPowerOn

- 功能

获取屏供电使能状态。

- 语法

```
MI_S32 MI_PANEL_GetPowerOn(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_PowerConfig_t *pstPowerCfg);
```

- 语法

表1-7

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstPowerCfg	该指针接收屏供电使能状态	输出

- 返回值

- 0 成功。
- 非0 失败，详情参照[错误码](#) [#jump]。

- 依赖

- 头文件：mi_panel.h、mi_panel_datatype.h
- 库文件：libmi_panel.a / libmi_panel.so

- 注意

- 目前不支持，建议app直接控制GPIO。

- 相关主题

[MI_PANEL_SetPowerOn](#) [#15-mi_panel_setpoweron]

1.7. MI_PANEL_SetBacklight

- 功能

设置屏背光使能。

- 语法

```
MI_S32 MI_PANEL_SetBacklight(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_BackLightConfig_t *pstBackLightCfg);
```

- 形参

表1-8

参数名称	描述	输入/输出
eIntfType	接口类型 输入	
pstBackLightCfg	屏背光使能状态	输入

- 返回值
 - 0 成功。
 - 非0 失败，[错误码 \[#jump\]](#)。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
- 注意
 - 目前不支持，建议app直接控制PWM。
- 相关主题
 - [MI_PANEL_GetBackLight](#) [#18-mi_panel_getbacklight]

1.8. MI_PANEL_GetBacklight

- 功能

获取屏背光使能状态。
- 语法

```
MI_S32 MI_PANEL_GetBacklight(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_BackLightConfig_t *pstBackLightCfg);
```

- 形参

表1-9

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstBackLightCfg	该指针接收屏背光使能状态	输出

- 返回值

- 0 成功。
- 非0 失败，[错误码](#) [#jump]。

- 依赖

- 头文件：mi_panel.h、mi_panel_datatype.h
- 库文件：libmi_panel.a / libmi_panel.so

- 注意

- 目前不支持，建议app直接控制PWM。

- 相关主题

[MI_PANEL_SetBacklight](#) [#17-mi_panel_setbacklight]

1.9. MI_PANEL_SetBacklightLevel

- 功能

设置背光亮度值。

- 语法

```
MI_S32 MI_PANEL_SetBacklightLevel(MI_PANEL_IntfType_e
eIntfType, MI_PANEL_BackLightConfig_t *pstBackLightCfg);
```

- 形参

表1-10

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstBackLightCfg	背光亮度值	输入

- 返回值
 - 0 成功。
 - 非0 失败，[错误码](#) [jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
- 注意
 - 调用前需先设置屏背光使能。
 - 目前不支持，建议app直接控制PWM。
- 相关主题

[MI_PANEL_GetBacklightLevel](#) [#110-mi_panel_getbacklightlevel]

1.10. MI_PANEL_GetBacklightLevel

- 功能

获取屏背光亮度值。
- 定义

```
MI_S32 MI_PANEL_GetBacklightLevel(MI_PANEL_IntfType_e  
eIntfType, MI_PANEL_BackLightConfig_t *pstBackLightCfg);
```

- 形参

表1-11

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstBackLightCfg	该指针接收屏背光亮度值	输出

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
- 注意
 - 调用前需要先设置背光亮度值。
 - 目前不支持，建议app直接控制PWM。

- 相关主题

[MI_PANEL_SetBacklightLevel](#) [#19-mi_panel_setbacklightlevel]

1.11. MI_PANEL_SetSscConfig

- 功能
设置展频相关信息。
- 定义

```
MI_S32 MI_PANEL_SetSscConfig(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_SscConfig_t *pstSscCfg);
```

- 形参

表1-12

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstSscCfg	展频相关配置信息	输入

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so

1.12. MI_PANEL_SetMipiDsiConfig

- 功能
设置MIPI接口相关配置
- 定义

```
MI_S32 MI_PANEL_SetMipiDsiConfig(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_MipiDsiConfig_t *pstMipiDsiCfg);
```

- 形参

表1-13

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstMipiDsiCfg	MIPI接口相关配置	输入

- 返回值

- 0 成功。
- 非0 失败，详情参照[错误码](#) [#jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so

- 注意

只有mipi panel才需要调用，同时要调用[MI_PANEL_SetPanelParam](#) [#116-mi_panel_setpanelparam]。

1.13. MI_PANEL_SetTimingConfig

- 功能
设置Timing相关信息
- 定义

```
MI_S32 MI_PANEL_SetTimingConfig(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_TimingConfig_t *pstTimingCfg);
```

- 形参

表1-14

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstTimingCfg	Timing相关配置信息	输入

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [#jump]。
- 依赖

- 头文件：mi_panel.h、mi_panel_datatype.h
- 库文件：libmi_panel.a / libmi_panel.so
- 相关主题

[MI_PANEL_GetTimingConfig](#) [#118-mi_panel_gettimingconfig]

1.14. MI_PANEL_SetDrvCurrentConfig

- 功能

设置驱动能力相关信息

- 定义

```
MI_S32 MI_PANEL_SetDrvCurrentConfig(MI_PANEL_IntfType_e
eIntfType, MI_PANEL_DrvCurrentConfig_t *pstDrvCurrentCfg);
```

- 形参

表1-15

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstDrvCurrentCfg	驱动能力相关配置信息	输入

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [#jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so

1.15. MI_PANEL_SetOutputPattern

- 功能
设置输出pattern
- 定义

```
MI_S32 MI_PANEL_SetOutputPattern(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_TestPatternConfig_t *pstTestPatternCfg);
```

- 形参

表1-16

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstTestPatternCfg	输出pattern相关配置信息	输入

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [#jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
- 注意
调试使用（目前只针对LVDS）

1.16. MI_PANEL_SetPanelParam

- 功能
设置PANEL参数
- 语法

```
MI_S32 MI_PANEL_SetPanelParam(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_ParamConfig_t *pstParamCfg);
```

- 形参

表1-17

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstParamCfg	屏参信息	输入

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [#jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
- 相关主题

[MI_PANEL_GetPanelParam](#) [#117-mi_panel_getpanelparam]

1.17. MI_PANEL_GetPanelParam

- 功能
获取屏参
- 定义

```
MI_S32 MI_PANEL_GetPanelParam(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_ParamConfig_t *pstParamCfg);
```

- 形参

表1-18

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstParamCfg	屏参信息	输出

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
- 注意

需先调用[MI_PANEL_SetPanelParam](#) [116-mi_panel_setpanelparam]。
- 相关主题

[MI_PANEL_SetPanelParam](#) [116-mi_panel_setpanelparam]

1.18. MI_PANEL_GetTimingConfig

- 功能

获取Timing相关信息
- 定义

```
MI_S32 MI_PANEL_GetTimingConfig(MI_PANEL_IntfType_e eIntfType,
MI_PANEL_TimingConfig_t *pstTimingCfg);
```

- 形参
- 表1-19

参数名称	描述	输入/输出
eIntfType	接口类型	输入
pstTimingCfg	Timing相关信息	输出

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so

- 相关主题

[MI_PANEL_SetTimingConfig](#) [#113-mi_panel_settimingconfig]

1.19. MI_PANEL_InitDev

- 功能
初始化PANEL设备
- 语法

```
MI_S32 MI_PANEL_InitDev(MI_PANEL_InitParam_t *pstInitParam);
```

- 形参

参数名称	描述	输入/输出
pstInitParam	设备初始化参数。	输入

- 返回值
 - 0 成功。

- 非0 失败，详情参照[错误码](#) [#jump]。
 - 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
 - 注意
 - 此接口在Version 2.09以上版本推荐使用，用于替换原有[MI_PANEL_InitDev](#) [#119-mi_panel_initdev]接口。
 - 相关主题[MI_PANEL_DeInitDev](#) [#120-mi_panel_deinitdev]
-

1.20. MI_PANEL_DeInitDev

- 功能
去初始化PANEL设备
- 语法

```
MI_S32 MI_PANEL_DeInitDev(void);
```

- 返回值
 - 0 成功。
 - 非0 失败，详情参照[错误码](#) [#jump]。
- 依赖
 - 头文件：mi_panel.h、mi_panel_datatype.h
 - 库文件：libmi_panel.a / libmi_panel.so
- 注意
 - 此接口在Version 2.09以上版本推荐使用，用于替换原有[MI_PANEL_DeInit](#)接口。
- 相关主题[MI_PANEL_InitDev](#) [#119-mi_panel_initdev]

2. PANEL数据类型

2.1. 数据类型定义

表2-1

数据类型	定义
MI_PANEL_PowerConfig_t [#22-mi_panel_powerconfig_t]	定义PANEL供电使能状态结构体
MI_PANEL_BackLightConfig_t [#23-mi_panel_backlightconfig_t]	定义PANEL背光信息结构体
MI_PANEL_SscConfig_t [#24-mi_panel_sscconfig_t]	定义PANEL展频配置信息结构体
MI_PANEL_MipiDsiConfig_t [#25-mi_panel_mipidsiconfig_t]	定义MIPI接口配置信息结构体
MI_PANEL_MipiDsiLaneMode_e [#26-mi_panel_mipidsilanemode_e]	定义MIPI接口Lane mode枚举
MI_PANEL_MipiDsiFormat_e [#27-mi_panel_mipidsiformat_e]	定义MIPI接口Format枚举
MI_PANEL_MipiDsiCtrlMode_e [#28-mi_panel_mipidsictrlmode_e]	定义MIPI接口CtrlMode枚举
MI_PANEL_TimingConfig_t [#29-mi_panel_timingconfig_t]	定义PANEL Timing信息结构体
MI_PANEL_DrvCurrentConfig_t [#210-mi_panel_drvcurrentconfig_t]	定义驱动能力配置结构体
MI_PANEL_TestPatternConfig_t [#211-mi_panel_testpatternconfig_t]	定义输出pattern信息结构体
数据类型 MI_PANEL_ParamConfig_t [#212-	定义 定义屏参信息结构体

mi_panel_paramconfig_t]		
MI_PANEL_IntfType_e	[#213-mi_panel_intftype_e]	定义接口类型枚举
MI_PANEL_AspectRatio_e	[#214-mi_panel_aspectratio_e]	定义AspectRatio枚举
MI_PANEL_TiBitMode_e	[#215-mi_panel_tibitmode_e]	定义TiBitMode枚举
MI_PANEL_OutputFormatBitMode_e	[#216-mi_panel_outputformatbitmode_e]	定义输出FormatBitMode枚举
MI_PANEL_OutputTimingMode_e	[#217-mi_panel_oututtimingmode_e]	定义panel output timing change mode枚举
MI_PANEL_InitParam_t	[#218-mi_panel_initparam_t]	定义panel设备初始化参数

注：本节已涵盖各重要的数据类型，部分未列出数据类型请参见 **mi_panel_datatype.h**

2.2. MI_PANEL_PowerConfig_t

- 说明
定义PANEL供电使能状态结构体
- 定义

```
typedef struct
{
    MI_BOOL bEn;
}MI_PANEL_PowerConfig_t;
```

- 相关数据类型及接口
[MI_PANEL_SetPowerOn](#) [#15-mi_panel_setpoweron]
[MI_PANEL_GetPowerOn](#) [#16-mi_panel_getpoweron]

2.3. MI_PANEL_BackLightConfig_t

- 说明

定义PANEL背光信息结构体

- 定义

```
typedef struct
{
    MI_BOOL bEn;

    MI_U16 u16Level;
}MI_PANEL_BackLightConfig_t;
```

- 成员

表2-2

成员名称	描述
bEn	是否使能panel背光
u16Level	如果bEn = TRUE，该参数设置背光level

- 相关数据类型及接口

- [MI_PANEL_SetBacklight](#) [#17-mi_panel_setbacklight]
- [MI_PANEL_GetBacklight](#) [#18-mi_panel_getbacklight]
- [MI_PANEL_SetBacklightLevel](#) [#19-mi_panel_setbacklightlevel]
- [MI_PANEL_GetBacklightLevel](#) [#110-mi_panel_getbacklightlevel]

2.4. MI_PANEL_SscConfig_t

- 说明

定义PANEL展频配置信息结构体

- 定义

```
typedef struct
{
    MI_BOOL bEn;

    MI_U16 u16Freq;

    MI_U16 u16Ratio;
}MI_PANEL_SscConfig_t;
```

- 成员

表2-3

成员名称	描述
bEn	panel展频功能使能状态
u16Freq	展频频率（单位Khz）
u16Ratio	展频幅度（单位1/1000）

- 注意

u16Freq不能小于15Khz.

- 相关数据类型及接口

[MI_PANEL_SetSscConfig](#) [#111-mi_panel_setsscconfig]

2.5. MI_PANEL_MipiDsiConfig_t

- 说明

定义MIPI接口配置信息结构体

- 定义

```
typedef struct
{
    MI_U8 u8HsTrail;

    MI_U8 u8HsPrpr;

    MI_U8 u8HsZero;

    MI_U8 u8ClkHsPrpr;

    MI_U8 u8ClkHsExit;

    MI_U8 u8ClkTrail;

    MI_U8 u8ClkZero;

    MI_U8 u8ClkHsPost;

    MI_U8 u8DaHsExit;

    MI_U8 u8ContDet;

    MI_U8 u8Lpx;

    MI_U8 u8TaGet;

    MI_U8 u8TaSure;

    MI_U8 u8TaGo;

    MI_U16 u16Hactive;

    MI_U16 u16Hpw;

    MI_U16 u16Hbp;

    MI_U16 u16Hfp;

    MI_U16 u16Vactive;

    MI_U16 u16Vpw;

    MI_U16 u16Vbp;

    MI_U16 u16Vfp;
```

```

MI_U16 u16Bllp;

MI_U16 u16Fps;

MI_PANEL_MipiDsiLaneMode_e enLaneNum;

MI_PANEL_MipiDsiFormat_e enformat;

MI_PANEL_MipiDsiCtrlMode_e enCtrl;

MI_U8 *pu8CmdBuf;

MI_U32 u32CmdBufSize;

MI_U8 u8SyncCalibrate;

MI_U16 u16VirHsyncSt;

MI_U16 u16VirHsyncEnd;

MI_U16 u16VsyncRef;

MI_U16 u16DataClkSkew;

MI_U8 u8PolCh0;

MI_U8 u8PolCh1;

MI_U8 u8PolCh2;

MI_U8 u8PolCh3;

MI_U8 u8PolCh4;

}MI_PANEL_MipiDsiConfig_t;

```

- 相关数据类型及接口

[MI_PANEL_MipiDsiLaneMode_e](#) [#26-mi_panel_mipidsilanemode_e]

[MI_PANEL_MipiDsiFormat_e](#) [#27-mi_panel_mipidsiformat_e]

[MI_PANEL_MipiDsiCtrlMode_e](#) [#28-mi_panel_mipidsictrlmode_e]

[MI_PANEL_SetMipiDsiConfig](#) [#112-mi_panel_setmipidsiconfig]

2.6. MI_PANEL_MipiDsiLaneMode_e

- 说明

定义MIPI接口Lane mode枚举

- 定义

```
typedef enum
{
    E_MI_PNL_MIPI_DSI_LANE_NONE = 0,
    E_MI_PNL_MIPI_DSI_LANE_1 = 1,
    E_MI_PNL_MIPI_DSI_LANE_2 = 2,
    E_MI_PNL_MIPI_DSI_LANE_3 = 3,
    E_MI_PNL_MIPI_DSI_LANE_4 = 4,
} MI_PANEL_MipiDsiLaneMode_e;
```

- 相关数据类型及接口

[MI_PANEL_MipiDsiConfig_t](#) [#25-mi_panel_mipidsiconfig_t]

[MI_PANEL_SetMipiDsiConfig](#) [#112-mi_panel_setmipidsiconfig]

2.7. MI_PANEL_MipiDsiFormat_e

- 说明

定义MIPI接口Format枚举

- 定义

```
typedef enum
{
    E_MI_PNL_MIPI_DSI_RGB565 = 0,
    E_MI_PNL_MIPI_DSI_RGB666 = 1,
```

```

E_MI_PNL_MIPI_DSI_LOOSELY_RGB666 = 2,

E_MI_PNL_MIPI_DSI_RGB888 = 3,

} MI_PANEL_MipiDsiFormat_e;

```

- 相关数据类型及接口

[MI_PANEL_MipiDsiConfig_t](#) [#25-mi_panel_mipidsiconfig_t]

[MI_PANEL_SetMipiDsiConfig](#) [#112-mi_panel_setmipidsiconfig]

2.8. MI_PANEL_MipiDsiCtrlMode_e

- 说明

定义MIPI接口CtrlMode枚举

- 定义

```

typedef enum

{

    E_MI_PNL_MIPI_DSI_CTRL_CMD_MODE = 0,

    E_MI_PNL_MIPI_DSI_CTRL_SYNC_PULSE = 1,

    E_MI_PNL_MIPI_DSI_CTRL_SYNC_EVENT = 2,

    E_MI_PNL_MIPI_DSI_BURST_MODE = 3,

} MI_PANEL_MipiDsiCtrlMode_e;

```

- 相关数据类型及接口

[MI_PANEL_MipiDsiConfig_t](#) [#25-mi_panel_mipidsiconfig_t]

[MI_PANEL_SetMipiDsiConfig](#) [#112-mi_panel_setmipidsiconfig]

2.9. MI_PANEL_TimingConfig_t

- 说明

定义PANEL Timing信息结构体

- 定义

```
typedef struct
{
    MI_U16 u16HSyncWidth;
    MI_U16 u16HSyncBackPorch;
    MI_U16 u16HSyncFrontPorch;
    MI_U16 u16VSyncWidth;
    MI_U16 u16VSyncBackPorch;
    MI_U16 u16VSyncFrontPorch;
    MI_U16 u16HStart;
    MI_U16 u16VStart;
    MI_U16 u16Height;
    MI_U16 u16Width;
    MI_U16 u16Htotal;
    MI_U16 u16Vtotal;
    MI_U16 u16Dclk;
}MI_PANEL_TimingConfig_t;
```

- 相关数据类型及接口

[MI_PANEL_SetTimingConfig](#) [#113-mi_panel_settimingconfig]

[MI_PANEL_GetTimingConfig](#) [#118-mi_panel_gettimingconfig]

2.10. MI_PANEL_DrvCurrentConfig_t

- 说明

定义驱动能力配置结构体

- 定义

```
typedef struct
{
    MI_U16 u16DrvCurrent;
}MI_PANEL_DrvCurrentConfig_t;
```

- 相关数据类型及接口

[MI_PANEL_SetDrvCurrentConfig](#) [#114-mi_panel_setdrvcurrentconfig]

2.11. MI_PANEL_TestPatternConfig_t

- 说明

定义输出pattern结构体

- 定义

```
typedef struct
{
    MI_BOOL bEn;

    MI_U16 u16R;

    MI_U16 u16G;

    MI_U16 u16B;
}MI_PANEL_TestPatternConfig_t;
```

- 相关数据类型及接口

[MI_PANEL_SetOutputPattern](#) [#115-mi_panel_setoutputpattern]

2.12. MI_PANEL_ParamConfig_t

- 说明

定义屏参结构体

- 定义

```
typedef struct
{
    const char *pPanelName; ///< PanelName

    MI_U8 u8Dither; ///< PANEL_DITHER, keep the setting

    MI_PANEL_IntfType_e eIntfType; ///< PANEL_LINK

    //////////////////////////////////////

    // Board related setting

    //////////////////////////////////////

    MI_U8 u8DualPort; ///< VOP_21[8], MOD_4A[1],
    PANEL_DUAL_PORT, refer to
    u8DoubleClk

    MI_U8 u8SwapPort; ///< MOD_4A[0], PANEL_SWAP_PORT, refer to
    "LVDS output
    app note" A/B channel swap

    MI_U8 u8SwapOdd_ML; ///< PANEL_SWAP_ODD_ML

    MI_U8 u8SwapEven_ML; ///< PANEL_SWAP_EVEN_ML

    MI_U8 u8SwapOdd_RB; ///< PANEL_SWAP_ODD_RB

    MI_U8 u8SwapEven_RB; ///< PANEL_SWAP_EVEN_RB

    MI_U8 u8SwapLVDS_POL; ///< MOD_40[5], PANEL_SWAP_LVDS_POL,
    for differential
    P/N swap

    MI_U8 u8SwapLVDS_CH; ///< MOD_40[6], PANEL_SWAP_LVDS_CH,
    for pair swap
```

```

MI_U8 u8PDP10BIT; ///< MOD_40[3], PANEL_PDP_10BIT ,for pair
swap

MI_U8 u8LVDS_TI_MODE; ///< MOD_40[2], PANEL_LVDS_TI_MODE,
refer to "LVDS
output app note"

////////////////////////////////////

// For TTL Only

////////////////////////////////////

MI_U8 u8DCLKDelay; ///< PANEL_DCLK_DELAY

MI_U8 u8InvDCLK; ///< MOD_4A[4], PANEL_INV_DCLK

MI_U8 u8InvDE; ///< MOD_4A[2], PANEL_INV_DE

MI_U8 u8InvHSync; ///< MOD_4A[12], PANEL_INV_HSYNC

MI_U8 u8InvVSync; ///< MOD_4A[3], PANEL_INV_VSYNC

////////////////////////////////////

// Output driving current setting

////////////////////////////////////

// driving current setting (0x00=4mA, 0x01=6mA, 0x02=8mA,
0x03=12mA)

MI_U8 u8DCKLCurrent; ///< define PANEL_DCLK_CURRENT

MI_U8 u8DECurrent; ///< define PANEL_DE_CURRENT

MI_U8 u8ODDDataCurrent; ///< define PANEL_ODD_DATA_CURRENT

MI_U8 u8EvenDataCurrent; ///< define
PANEL_EVEN_DATA_CURRENT

////////////////////////////////////

// panel on/off timing

////////////////////////////////////

MI_U16 u16OnTiming1; ///< time between panel & data while
turn on power

```

```

    MI_U16 u16OnTiming2; ///< time between data & back light
while turn on
    power

    MI_U16 u16OffTiming1; ///< time between back light & data
while turn off
    power

    MI_U16 u16OffTiming2; ///< time between data & panel while
turn off power

    //////////////////////////////////////

    // panel timing spec.

    //////////////////////////////////////

    // sync related

    MI_U16 u16HSyncWidth; ///< VOP_01[7:0], PANEL_HSYNC_WIDTH

    MI_U16 u16HSyncBackPorch; ///< PANEL_HSYNC_BACK_PORCH, no
register setting,
    provide value for query only

    MI_U16 u16VSyncWidth; ///< define PANEL_VSYNC_WIDTH

    MI_U16 u16VSyncBackPorch; ///< define
PANEL_VSYNC_BACK_PORCH

    // DE related

    MI_U16 u16HStart; ///< VOP_04[11:0], PANEL_HSTART, DE H
Start
    (PANEL_HSYNC_WIDTH + PANEL_HSYNC_BACK_PORCH)

    MI_U16 u16VStart; ///< VOP_06[11:0], PANEL_VSTART, DE V
Start

    MI_U16 u16Width; ///< PANEL_WIDTH, DE width (VOP_05[11:0] =
HEnd = HStart +
    Width - 1)

    MI_U16 u16Height; ///< PANEL_HEIGHT, DE height
(VOP_07[11:0], = Vend =
    VStart + Height - 1)

    // DClk related

```

```

    MI_U16 u16MaxHTotal; ///< PANEL_MAX_HTOTAL. Reserved for
future using.

    MI_U16 u16HTotal; ///< VOP_0C[11:0], PANEL_HTOTAL

    MI_U16 u16MinHTotal; ///< PANEL_MIN_HTOTAL. Reserved for
future using.

    MI_U16 u16MaxVTotal; ///< PANEL_MAX_VTOTAL. Reserved for
future using.

    MI_U16 u16VTotal; ///< VOP_0D[11:0], PANEL_VTOTAL

    MI_U16 u16MinVTotal; ///< PANEL_MIN_VTOTAL. Reserved for
future using.

    MI_U16 u16MaxDCLK; ///< PANEL_MAX_DCLK. Reserved for future
using.

    MI_U16 u16DCLK; ///< LPLL_0F[23:0], PANEL_DCLK

    MI_U16 u16MinDCLK; ///< PANEL_MIN_DCLK. Reserved for future
using.

    ///< spread spectrum

    MI_U16 u16SpreadSpectrumStep; ///< move to board define, no
use now.

    MI_U16 u16SpreadSpectrumSpan; ///< move to board define, no
use now.

    MI_U8 u8DimmingCtl; ///< Initial Dimming Value

    MI_U8 u8MaxPWMVal; ///< Max Dimming Value

    MI_U8 u8MinPWMVal; ///< Min Dimming Value

    MI_U8 u8DeinterMode; ///< define PANEL_DEINTER_MODE, no use
now

    MI_PANEL_AspectRatio_e ePanelAspectRatio; ///< Panel Aspect
Ratio, provide
    information to upper layer application for aspect ratio
setting.

    MI_U16 u16LVDSTxSwapValue;

```



```

MI_PANEL_TiBitMode_e eTiBitMode; ///< MOD_4B[1:0], refer to
"LVDS output
app note"

MI_PANEL_OutputFormatBitMode_e eOutputFormatBitMode;

MI_U8 u8SwapOdd_RG; ///< define PANEL_SWAP_ODD_RG

MI_U8 u8SwapEven_RG; ///< define PANEL_SWAP_EVEN_RG

MI_U8 u8SwapOdd_GB; ///< define PANEL_SWAP_ODD_GB

MI_U8 u8SwapEven_GB; ///< define PANEL_SWAP_EVEN_GB

/**

* Others

*/

MI_U8 u8DoubleClk; ///< LPLL_03[7], define Double Clock
,LVDS dual mode

MI_U32 u32MaxSET; ///< define PANEL_MAX_SET

MI_U32 u32MinSET; ///< define PANEL_MIN_SET

MI_PANEL_OutputTimingMode_e eOutTimingMode; ///

```

- 相关数据类型及接口

[MI_PANEL_IntfType_e](#) [#216-mi_panel_intfType_e]

[MI_PANEL_AspectRatio_e](#) [#214-mi_panel_aspectratio_e]

[MI_PANEL_TiBitMode_e](#) [#215-mi_panel_tibitmode_e]

[MI_PANEL_OutputFormatBitMode_e](#) [#216-
mi_panel_outputformatbitmode_e]

[MI_PANEL_OutputTimingMode_e](#) [#217-mi_panel_outputtimingmode_e]

2.13. MI_PANEL_IntfType_e

- 说明
定义PANEL Interface Type枚举
- 定义

```
typedef enum
{
    E_MI_PNL_INTF_TTL,                ///< TTL
    type
    E_MI_PNL_INTF_LVDS,                ///< LVDS
    type
    E_MI_PNL_INTF_RSDS,                ///< RSDS
    type
    E_MI_PNL_INTF_MINILVDS,            ///< TCON
    E_MI_PNL_INTF_ANALOG_MINILVDS,     ///< Analog
    TCON
    E_MI_PNL_INTF_DIGITAL_MINILVDS,    ///<
    Digital TCON
    E_MI_PNL_INTF_MFC,                ///< Ursa
    (TTL output to Ursa)
    E_MI_PNL_INTF_DAC_I,               ///< DAC output
    E_MI_PNL_INTF_DAC_P,               ///< DAC output
    E_MI_PNL_INTF_PDPLVDS,             ///< For
    PDP(Vsync use Manually MODE)
    E_MI_PNL_INTF_EXT,                ///< EXT
    LPLL TYPE
    E_MI_PNL_INTF_MIPI_DSI,            ///< Mipi
    DSI
```

```

        E_MI_PNL_INTF_BT656,                //BT656
interface

        E_MI_PNL_INTF_BT601,                ///< BT601

        E_MI_PNL_INTF_BT1120,               ///< BT1120

        E_MI_PNL_INTF_MCU_TYPE,              ///< MCU
Type

        E_MI_PNL_INTF_SRGB,                  ///< sRGB

        E_MI_PNL_INTF_TTL_SPI_IF,           ///< TTL
with SPI init interface

}MI_PANEL_IntfType_e;

```

- 相关数据类型及接口

[MI_PANEL_ParamConfig_t](#) [#212-mi_panel_paramconfig_t]

[MI_PANEL_GetPanelParam](#) [#117-mi_panel_getpanelparam]

2.14. MI_PANEL_AspectRatio_e

- 说明

定义PANEL AspectRatio枚举

- 定义

```

typedef enum

{

    E_MI_PNL_ASPECT_RATIO_4_3 = 0, ///< set aspect ratio to 4 :
3

    E_MI_PNL_ASPECT_RATIO_WIDE, ///< set aspect ratio to 16 : 9

    E_MI_PNL_ASPECT_RATIO_OTHER, ///< resvered for other aspect
ratio other
    than 4:3/ 16:9

}MI_PANEL_AspectRatio_e;

```

- 相关数据类型及接口

[MI_PANEL_ParamConfig_t](#) [#212-mi_panel_paramconfig_t]

[MI_PANEL_GetPanelParam](#) [#117-mi_panel_getpanelparam]

2.15. MI_PANEL_TiBitMode_e

- 说明

定义PANEL TiBitMode枚举

- 定义

```
typedef enum
{
    E_MI_PNL_TI_10BIT_MODE = 0,
    E_MI_PNL_TI_8BIT_MODE = 2,
    E_MI_PNL_TI_6BIT_MODE = 3,
} MI_PANEL_TiBitMode_e;
```

- 相关数据类型及接口

[MI_PANEL_ParamConfig_t](#) [#212-mi_panel_paramconfig_t]

[MI_PANEL_GetPanelParam](#) [#117-mi_panel_getpanelparam]

2.16. MI_PANEL_OutputFormatBitMode_e

- 说明

定义PANEL OutputFormatBitMode枚举

- 定义

```
typedef enum
```

```

{

    E_MI_PNL_OUTPUT_10BIT_MODE = 0, //default is 10bit, because
    8bit panel can
    use 10bit config and 8bit config.

    E_MI_PNL_OUTPUT_6BIT_MODE = 1, //but 10bit panel(like PDP
    panel) can only
    use 10bit config.

    E_MI_PNL_OUTPUT_8BIT_MODE = 2, //and some PDA panel is
    6bit.

    E_MI_PNL_OUTPUT_565BIT_MODE = 3,

} MI_PANEL_OutputFormatBitMode_e;

```

- 相关数据类型及接口

[MI_PANEL_ParamConfig_t](#) [#212-mi_panel_paramconfig_t]

[MI_PANEL_GetPanelParam](#) [#117-mi_panel_getpanelparam]

2.17. MI_PANEL_OutputTimingMode_e

- 说明

定义panel output timing change mode

- 定义

```

typedef enum

{

    E_MI_PNL_CHG_DCLK = 0, ///<change output Dclk to change
    Vfreq.

    E_MI_PNL_CHG_HTOTAL = 1, ///<change H total to change
    Vfreq.

    E_MI_PNL_CHG_VTOTAL = 2, ///<change V total to change
    Vfreq.

} MI_PANEL_OutputTimingMode_e;

```

- 相关数据类型及接口

[MI_PANEL_ParamConfig_t](#) [#212-mi_panel_paramconfig_t]

[MI_PANEL_GetPanelParam](#) [#117-mi_panel_getpanelparam]

2.18. MI_PANEL_InitParam_t

- 说明

定义panel设备初始化参数

- 定义

```
typedef struct MI_PANEL_InitParam_s
{
    MI_PANEL_IntfType_e eIntfType ;    // PANEL interface type
    enumeration
} MI_PANEL_InitParam_t;
```

- Related data types and interfaces

[MI_PANEL_InitDev](#) [#119-mi_panel_initdev]

3. PANEL error code

Table 3-1

error code	macro definition	describe
0xA00B2011	MI_ERR_PANEL_INVALID_PARAM	illegal parameter
0xA00B2006	MI_ERR_PANEL_NULL_PTR	null pointer exception
0xA00B2001	MI_ERR_PANEL_INVALID_DEVID	invalid device id
0xA00B2018	MI_ERR_PANEL_DRV_FAILED	PANEL driver returns failure
0xA00B2015	MI_ERR_PANEL_NOT_INIT	PANEL not initialized
0xA00B2008	MI_ERR_PANEL_NOT_SUPPORT	Operation not supported